

AI Roadmap 2026: Python → Generative AI → Agentic AI

Presented by:
Kodi Prakash Senapati
Lead Data Architect

Agenda

- Introduction
- Python Foundations
- ML & DL
- Generative AI
- Agentic AI
- 2026 Roadmap
- Q/A

Why AI Skills Matter in 2026

Automation → Intelligence → Autonomy
AI transforms every industry.
Skills required: Python → Gen AI → Agents

Python Foundations

- Python basics
- Data types, loops, functions
- Numpy & Pandas
- Matplotlib
- Data pipelines

Data Analysis Skills

- Data cleaning
- Exploratory analysis
- Insight extraction
- Business dashboards

Machine Learning

- Regression
- Classification
- Clustering
- Scikit-learn
- Model evaluation

Deep Learning Foundations

- ANN • CNN • RNN • LSTM
- PyTorch & TensorFlow
- Why DL leads to Gen AI

Transformers & Modern AI

- Attention mechanism
- GPT, BERT, LLaMA

- Multimodal AI

Generative AI Overview

- LLMs
- Tokenization
- Embeddings
- Prompt engineering
- LoRA Fine-tuning

Gen AI Tools

- Hugging Face
- LangChain
- OpenAI API
- Stable Diffusion

Agentic AI (2026)

AI that can:

- Think
- Plan
- Act
- Execute tasks autonomously

Agent Capabilities

- Tool usage
- Multi-step reasoning
- API calling
- Web browsing
- Autonomous workflows

2026 AI Roadmap

1. Python
2. ML
3. Deep Learning
4. Generative AI
5. Agents
6. Projects

Industry Use Cases

- Finance AI
- Healthcare AI
- Retail Analytics
- Automated Enterprise Agents

Thank You

Presented by:
Kodi Prakash Senapati
Lead Data Architect